

## 32 | OVERVIEW



## Examining Past Projects in Climate, Health, and Energy

### Abstract

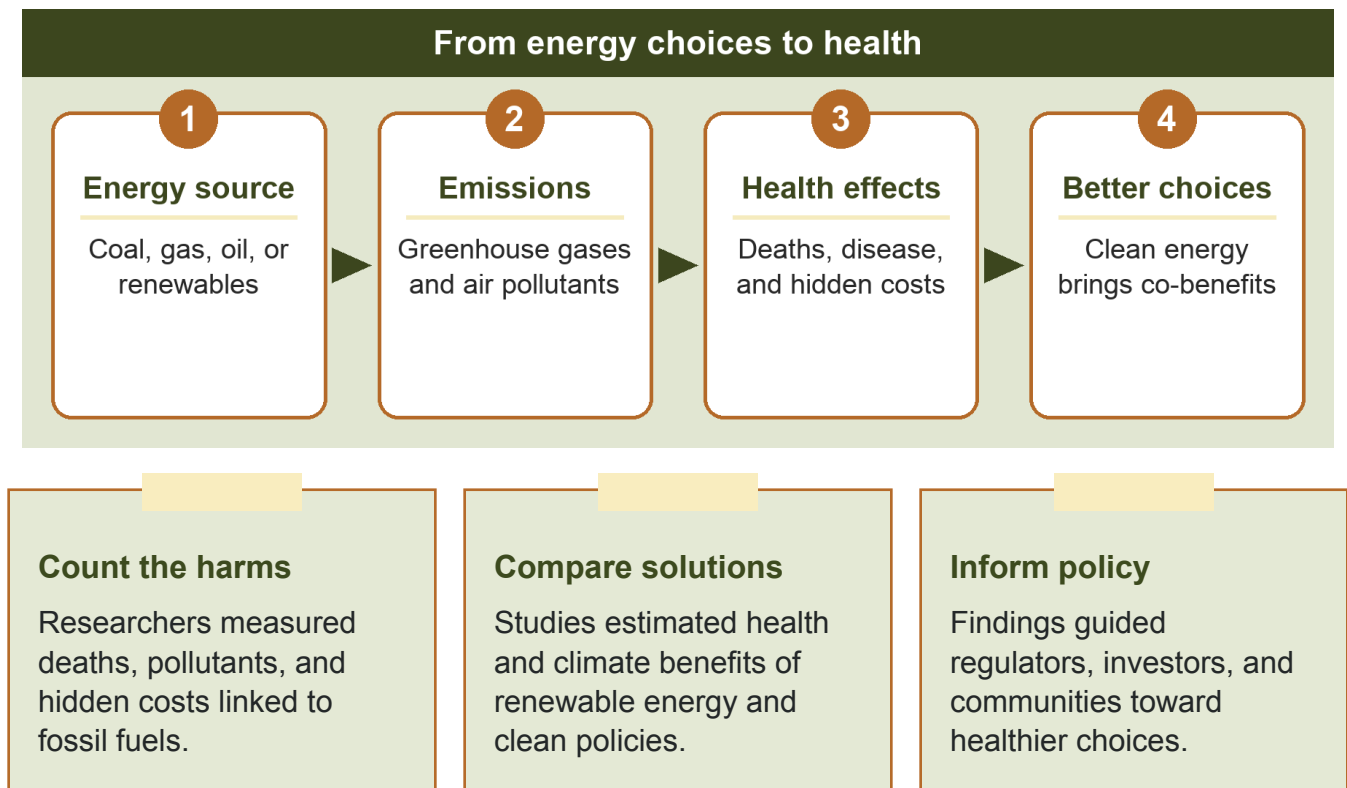
For over 25 years, the Harvard Chan Center for Climate, Health, and the Global Environment has produced research connecting energy, climate, and human health. This lesson surveys its past projects, from finding that fossil fuel air pollution causes 1 in 5 deaths worldwide to measuring pollutants in gas stoves and calculating how cheaply pandemics could be prevented, and asks what they reveal about the health case for climate action.

### Introduction

How we produce energy shapes both the climate and our health. Burning fossil fuels releases greenhouse gases that warm the planet, but it also creates air pollution that harms our lungs and hearts today. Destroying forests and changing how land is used can push animals and people closer together, raising the risk that new diseases spill over. Over more than 25 years, Harvard Chan C-CHANGE has studied these connections through projects on infectious disease, fossil fuels, natural gas, clean energy, and climate policy. Looking back at these projects shows how research builds a case that protecting the climate is also one of the most powerful ways to protect health.

## How Do Scientists Study This?

C-CHANGE's past projects used a range of methods. Literature reviews summarized a decade of research on methane and air pollution from the oil and gas industry. Modeling studies estimated deaths from fossil fuel air pollution worldwide. Full cost accounting added up the hidden costs of coal across its entire life cycle, from mining to burning. Field studies tested for health-damaging pollutants in unburned natural gas in people's homes and mapped how many people live near underground gas storage wells. Benefit analyses estimated the lives and dollars saved by offshore wind, clean electricity standards, and power plant carbon rules. Economic studies compared the cost of preventing pandemics with the cost of lives lost.



## What Did They Discover?

The projects produced striking findings. Deaths from fossil fuel air pollution were higher than previously thought: more than 8 million people per year, or 1 in 5 deaths worldwide. The life cycle of coal was found to cost the U.S. public a third to over half a trillion dollars each year. A study of natural gas in homes was the first to test for health-damaging pollutants in unburned gas, and another found about 53,000 people live within a city block of underground gas storage wells. On the solutions side, preventing the next pandemic was estimated to cost just 2% of what COVID-19 cost, and offshore wind and clean electricity standards could save many lives and billions of dollars.

### Past projects by the numbers

**8M+**

deaths per year from fossil fuel air pollution

**1 in 5**

deaths worldwide linked to fossil fuel pollution

**2%**

cost of preventing a pandemic versus COVID-19

Many studies highlight co-benefits: cutting carbon dioxide from power plants also cuts air pollutants, bringing near-term health gains.

## Takeaways

### Fossil fuels harm health

Air pollution from fossil fuels causes about 1 in 5 deaths worldwide.

### Hidden costs are huge

Coal's full life cycle costs the U.S. public hundreds of billions of dollars a year.

### Prevention pays

Clean energy and pandemic prevention save lives and money.

## Key Terms

### Full cost accounting

Adding up all the costs of a product or activity, including hidden health and environmental costs, across its life cycle.

### Life cycle

All the stages of a product, from extraction and production to use and disposal.

### Spillover

When a pathogen jumps from animals to people, which can start a new disease outbreak.

### Co-benefits

Extra positive effects of an action, such as health gains from cutting greenhouse gas emissions.

### Clean electricity standard

A policy requiring a set share of electricity to come from low- or zero-carbon sources by a target date.

#### Match-Up: draw a line from each term to its clue

**A** Full cost accounting

**B** Life cycle

**C** Spillover

**D** Co-benefits

**E** Clean electricity standard

**1** When a pathogen jumps from animals to people, which can start a...

**2** Extra positive effects of an action

**3** A policy requiring a set share of electricity to come from low-...

**4** All the stages of a product, from extraction and production to use...

**5** Adding up all the costs of a product or activity, including hidden...

Answer key: A-5 · B-4 · C-1 · D-2 · E-3

## Activity

### Research Gallery Walk

Students explore past C-CHANGE projects and build a case for the health benefits of climate action.

Materials: Five station posters: infectious disease, fossil fuels, natural gas, clean energy, climate policy; Sticky notes; Evidence organizer

1. Visit each station and record one key finding with its number, such as 1 in 5 deaths.
2. At each station, write on a sticky note who would benefit most from acting on that finding.
3. Back in groups, rank the five findings by how convincing they are for a city council.
4. Write a short speech using three findings to argue that climate action is health action.
5. Discuss: why might research on costs and lives saved persuade people who are not focused on climate?

## References

Harvard T.H. Chan School of Public Health, C-CHANGE. Past Projects.  
<https://hsph.harvard.edu/research/climate-health-c-change/research/past-projects/>